STAT 5703 **Recitation 5** Spring 2026

Solutions

1. Let $X_1, X_2, \dots, X_n \sim N(\mu, 1)$ be and iid sample, where *only* the mean μ is unknown. Consider

$$H_0: \mu = 0 \text{ vs. } H_a: \mu = 1$$

Let the rejection region be $R = \{X_1, \dots, X_n : \bar{X} > c\}$ based on a significance level α.

a) Find the formula for c as a function of n and α.

Under the assumption of H_0: $\mu = 0$, we know that $\bar{X} \sim N\left(0, \frac{1}{\sqrt{n}}\right)$

Therefore,

$$P(\bar{X} > c | \mu = 0) = P\left(Z > \sqrt{n}c\right) = 1 - \Phi\left(\sqrt{n}c\right)$$
$$\Rightarrow \alpha = 1 - \Phi\left(\sqrt{n}c\right) \iff \Phi\left(\sqrt{n}c\right) = 1 - \alpha$$
$$\iff \sqrt{n}c = \Phi^{-1}(1-\alpha) \iff c = \frac{\Phi^{-1}(1-\alpha)}{\sqrt{n}}$$

Note that $\Phi^{-1}(1-\alpha)$ is just the upper α quantile, z_α, of the standard normal.

b) Find the power of the test π(1) as function of c and n.

$$\pi(1) = P(\bar{X} > c | \mu = 1) = P\left(Z > \sqrt{n}(c-1)\right) = 1 - \Phi\left(\sqrt{n}(c-1)\right)$$
$$= 1 - \Phi\left(\sqrt{n}\left(\frac{\Phi^{-1}(1-\alpha)}{\sqrt{n}} - 1\right)\right) = 1 - \Phi\left(\Phi^{-1}(1-\alpha) - \sqrt{n}\right)$$

c) Show theoretically that π (1) → 1 as $n \to \infty$. (Note this is a sequence of real numbers.)

When $n \to \infty$, so does $\sqrt{n} \to \infty$, therefore $\Phi^{-1}(1-\alpha) - \sqrt{n} \to -\infty$, which means that

$$\pi(1) \to 1 - \Phi(-\infty) = 1 - 0 = 1$$

d) Verify by simulation your answers to parts a) and b). Choose α = 0.03, $n = 80$ and simulate R = 100,000 samples $X_1, X_2, \dots, X_n \sim N(\mu, 1)$, where $\mu = 0$ for part a) and $\mu = 1$ for part b). For part a) compute how many times you rejected the correct H_0 and for part b) how many times you rejected the wrong H_0. For part a) this proportion should match approximately with α and for part b) the proportion should match your formula.

```python
import numpy as np
from scipy.stats import norm

# Given values
alpha = 0.03
n = 80
R = 100000
mu_a = 0  # for part a)
mu_b = 1  # for part b)
```

```python
Z_alpha = norm.ppf(1 - alpha)
c = Z_alpha / np.sqrt(n)
```

```python
# Simulation for part a)
reject_a = 0
for _ in range(R):
    sample = np.random.normal(mu_a, 1, n)
    sample_mean = np.mean(sample)
    if sample_mean > c:
        reject_a += 1
```

```python
# Proportion of rejections for part a)
proportion_a = reject_a / R
```

```python
# Simulation for part b)
reject_b = 0
for _ in range(R):
    sample = np.random.normal(mu_b, 1, n)
    sample_mean = np.mean(sample)
    if sample_mean > c:
        reject_b += 1
```

```python
# Proportion of rejections for part b)
proportion_b = reject_b / R
```

```
]: print(f"Proportion of rejections for part a : {proportion_a}")
   print(f"Proportion of rejections for part b : {proportion_b}")

   Proportion of rejections for part a : 0.02941
   Proportion of rejections for part b : 1.0

]: theoretical_power = 1 - norm.cdf(np.sqrt(n) * (c - mu_b))
   print(theoretical_power)

   0.9999999999991881
```

2. Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f(x) = \theta x^{\theta-1}, \ 0 < x < 1$$

a) Find the RR of the most powerful test for testing

$$H_0: \theta = 3 \text{ vs. } H_a: \theta = 2$$

$$L(\theta) = \prod_{i=1}^{n}\left(\theta x_i^{\theta-1}\right) = \theta^n\left(\prod_{i=1}^{n} x_i\right)^{\theta-1}$$

$$\Rightarrow \frac{L(\theta_0)}{L(\theta_a)} = \frac{3^n(\prod_{i=1}^{n} x_i)^{3-1}}{2^n(\prod_{i=1}^{n} x_i)^{2-1}} = \left(\frac{3}{2}\right)^n \prod_{i=1}^{n} x_i$$

$$\Rightarrow \frac{L(\theta_0)}{L(\theta_a)} \le k \iff \prod_{i=1}^{n} x_i \le c$$

b) Find the critical value c if $n = 1$, $\alpha = 0.05$

$$P(X \le c | X \sim f(x) = 3x^2) = 0.05$$

$$\iff \int_0^c 3x^2 dx = 0.05 \iff c = \sqrt[3]{0.05}$$

3. Let $X_1, \dots, X_n \sim N(0, \sigma)$

Find the RR of the most powerful test (Neyman-Pearson method) for testing

$$H_0: \sigma^2 = \sigma^2_0 \text{ vs. } H_a: \sigma^2 = \sigma^2_a$$

where $\sigma^2_0 < \sigma^2_a$

$$L(\boldsymbol{\sigma^2}) = \prod_{i=1}^{n}\left[(2\pi)^{-\frac{1}{2}}(\sigma^2)^{-\frac{1}{2}}\mathrm{e}^{-\frac{x_i^2}{2\sigma^2}}\right]$$

$$= (2\pi)^{-\frac{n}{2}}(\sigma^2)^{-\frac{n}{2}}\mathrm{e}^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n} x_i^2}$$

$$\Rightarrow \frac{L(\sigma_0^2)}{L(\sigma_a^2)} = \frac{(2\pi)^{-\frac{n}{2}}(\sigma_0^2)^{-\frac{n}{2}}\mathrm{e}^{-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n} x_i^2}}{(2\pi)^{-\frac{n}{2}}(\sigma_a^2)^{-\frac{n}{2}}\mathrm{e}^{-\frac{1}{2\sigma_a^2}\sum_{i=1}^{n} x_i^2}} = (\sigma_a^2)^{\frac{n}{2}}(\sigma_0^2)^{-\frac{n}{2}}\mathrm{e}^{\left(\frac{1}{2\sigma_a^2}-\frac{1}{2\sigma_0^2}\right)\sum_{i=1}^{n} x_i^2} \le k$$

$$\iff \mathrm{e}^{\left(\frac{1}{2\sigma_a^2}-\frac{1}{2\sigma_0^2}\right)\sum_{i=1}^{n} x_i^2} \le k \iff \left(\frac{1}{2\sigma_a^2} - \frac{1}{2\sigma_0^2}\right)\sum_{i=1}^{n} x_i^2 \le k$$

Since $\frac{1}{\sigma_a^2} < \frac{1}{\sigma_0^2}$

$$R = \sum_{i=1}^{n} X_i^2 \ge c$$

Exercise: Find c when $n = 50, \alpha = 0.05, \sigma_0^2 = 1, \sigma_a^2 = 2$

4. A restaurant owner wants to test if her two managers perform at the same level. To do that, she collects data about the number of customer complaints at two random samples of shifts (one for each manager). Here is a summary of the results:

	Manager A	Manager B
Mean	4 complaints	5 complaints
Standard deviation	0.3	0.5
Number of shifts	19	21

Hint: Use (without deriving it) the two-sample *t*-test assuming equal population variances.

$$s_p = \sqrt{\frac{(m-1)s_x^2 + (n-1)s_y^2}{m+n-2}} = \sqrt{\frac{18(0.3)^2 + 20(0.5)^2}{38}} \approx 0.4$$

Test Statistic:

$$t = \frac{\bar{x} - \bar{y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}} = \frac{4-5}{0.4\sqrt{\frac{1}{19} + \frac{1}{21}}} \approx -7.9$$

Find the p-value and state the conclusion.

5. Let $X_1, \ldots, X_n \sim N(\mu, \sigma)$
Find the LRT rule for testing

$$H_0: \sigma^2 = \sigma^2_0 \text{ vs. } H_a: \sigma^2 \neq \sigma^2_0$$

Note that the parameter space is $\Theta = \mathbb{R} \times \mathbb{R}_+$

$$L(\mu, \sigma) = \prod_{i=1}^{n}\left[(2\pi\sigma^2)^{-½} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}\right]$$

MLEs have been derived before as $\hat{\mu}_{MLE} = \bar{x}, \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

The maximum likelihood under H_0 is:

$$\sup_{\mu\in\mathbb{R},\sigma=\sigma_0} L(\mu, \sigma) = L(\bar{x}, \sigma_0) = (2\pi)^{-\frac{n}{2}}(\sigma_0^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

$$\Rightarrow \Lambda(\boldsymbol{x}) = \frac{\sup_{\mu\in\mathbb{R},\sigma=\sigma_0} L(\mu, \sigma)}{\sup_{\mu\in\mathbb{R},\sigma>0} L(\mu, \sigma)} = \frac{(\sigma_0^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n}(x_i-\bar{x})^2}}{\left[\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2\right]^{-\frac{n}{2}} e^{-\frac{n}{2}}} \leq k$$

$$\Leftrightarrow \left[\frac{(n-1)s^2}{\sigma_0^2}\right]^{\frac{n}{2}} e^{-\frac{(n-1)s^2}{2\sigma_0^2}} \leq k_1$$

Notice that the lefthand side is a function $f(x) = x^{\frac{n}{2}} e^{-\frac{x}{2}}$, where $x = \frac{(n-1)s^2}{\sigma_0^2} > 0$. The function is unimodal (assuming $n > 1$), so the inequality $f(x) \leq k_1$ is equivalent to

$$\frac{(n-1)s^2}{\sigma_0^2} \leq c_1 \cup \frac{(n-1)s^2}{\sigma_0^2} \geq c_2$$

Extra (if you have time)

6. Suppose $\boldsymbol{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Define $\boldsymbol{W} = \begin{bmatrix} W_1 \\ W_2 \end{bmatrix} = \begin{bmatrix} X_1 - X_2 \\ X_2 - X_3 \end{bmatrix}$. Find the distribution of $\boldsymbol{W}$.

Answer: Bivariate normal with mean vector and covariance matrix ….